Alin-Petru Rosu

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Summary

Cryptography research engineer with a strong interest in PKI and secure space communications. Delivered satellite ground-segment cryptographic services and open-source libraries for privacy-enhancing technologies. Author of scientific papers, including work on space PKI. Motivated to explore cryptography and secure communications in critical infrastructure, with a drive for continuous learning.

Experience

Cryptography Research Software Engineer – Verifiable FHE 06/2026 – Present
IMDEA Software Institute *Madrid, Spain*

- **Stack**: *C, Python, CKKS, SNARKs, CPU Intrinsics*
- Designed and implemented an open-source library for verifiable **CKKS** homomorphic encryption with **SNARKs**.
- Optimised performance-critical polynomial arithmetic using **CPU intrinsics** and profiling-guided tuning.
- Developed a **machine learning PoC** for end-to-end verifiable inference over encrypted data.

Cybersecurity Software Engineer – Ground Segment Cryptography 08/2025 – 03/2026
European Space Agency *(via Telespazio Belgium)* *Noordwijk, The Netherlands*

- **Stack**: *Java, Scala, C++, JCE, Bouncy Castle, OpenSSL, PKCS#11, XMSS, EJBCA, Docker*
- Owned the design and implementation of a **ground segment service exposing** 27 **cryptographic functions** over UDP, including encryption, signatures, hashing, key derivation, PKI, and a hybrid key negotiation protocol.
- Implemented **key management for stateful hash-based signatures** using BKS keystores and HSM.
- Proposed and integrated a **self-hosted *EJBCA* PKI** via REST API, halving custom PKI development time.

Cybersecurity Intern – Public Key Infrastructure 10/2024 – 07/2025
European Space Agency *Noordwijk, The Netherlands*

- **Stack**: *C++, OpenSSL, ML-KEM, ML-DSA, X.509, DER, C509, CBOR*
- Researched **post-quantum certificate formats** for constrained and interoperable space PKI deployments.
- Designed the **first open-source tool for composite *C509***, supporting lattice-based cryptography with *ECC*.
- Demonstrated that *CBOR CRLs* reduce size by 55% at 1k entries, quantifying DER encoding overhead at scale.
- Proposed *SCVP* for federated space PKI validation, leading to an **invitation for in-orbit experimentation**.

Cybersecurity Research Assistant – Privacy-Preserving Blockchain 05/2024 – 09/2024
Delft University of Technology *Delft, The Netherlands*

- **Stack**: *Hyperledger Fabric, Java, Spring Boot, Protobuf, Docker*
- Coordinated a team of four and **took full technical ownership** of a platform for medical device data research.
- Designed a permissioned *Hyperledger Fabric* network with end-to-end TLS, delivering a **prototype in 4 weeks**.
- Implemented privacy-preserving smart-contract queries using *threshold Paillier homomorphic encryption*, *ε-differential privacy*, and *integer linear programming*, enabling grouped counting without per-record decryption.
- Received **Best Demo Award** and integrated into the EU-funded *SEPTON* initiative as a candidate architecture.

Backend Software Engineer – Payments Processing 08/2022 – 07/2023
ING Bank *Bucharest, Romania*

- Developed ISO 20022 instant payment systems in *Scala*, **sustaining 90+ transactions per second**.
- Delivered features and resolved bugs in a 10+ developer agile team, onboarding three engineers to full autonomy.

Software & DevOps Engineer 06/2020 – 07/2022
BusyMachines *Timisoara, Romania*

- **Owned cloud infrastructure and CI/CD** on AWS for a mentoring and videoconferencing platform.
- **Owned a 15+ node on-premises *Docker Swarm* platform** for bioinformatics processing.
- Developed backend services in *Java* and *Scala* across multiple client projects.

Education

Delft University of Technology — Delft, The Netherlands
MSc Computer Science, Cybersecurity (GPA: 8.7/10, Cum Laude) — *08/2023 – 07/2025*

- **Teaching Assistant**: Compiler Construction, Computer Security
- **Coursework**: Security and Cryptography, Privacy-Enhancing Technologies, Blockchain Engineering

Politehnica University of Timisoara — Timisoara, Romania
MEng Software Engineering (GPA: 9.5/10) — *08/2022 – 07/2023*

Politehnica University of Timisoara — Timisoara, Romania
BEng Computer and Information Technology (GPA: 9.66/10) — *08/2018 – 07/2022*

- **Project**: *jFamilyCounselor*, *Java* static analysis for hidden familial type correlations (**Best Paper Award**)

Projects

ers *(Encrypted Range Search)* | **Stack**: *Python* — 10/2024 – 01/2025

- Implemented encrypted range search schemes using Hilbert curves for multi-dimensional query processing.
- Evaluated algorithmic trade-offs via synthetic benchmarks, building practical intuition for applied cryptography.

Additional Information

Publications: *https://orcid.org/0009-0005-8537-7265*

Selected Presentations & Talks:

- Speaker, *CYSAT Madrid* — *10/2026*
- Panellist, *Romanian Space Initiative International* — *11/2025*
- Author, *Security for Space Systems Conference* — *11/2025*
- Presenter, *CCSDS Spring Meeting* — *05/2025*

Languages: English (C1), Spanish (C1), German (B1)